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Sheet conveyance device and image forming apparatus

Abstract

A transmission gear is movable in an axial direction between a first position separated from an output gear and a second position coupled to the output gear in a state where the transmission gear engages with an input gear. The input gear is configured to: by rotating in a first direction by receiving driving force from a drive source, move the transmission gear from the first position to the second position and transmit driving force in the first direction to the output gear via the transmission gear; and by rotating in a second direction, move the transmission gear from the second position to the first position and cancel transmission of driving force to the output gear. The transmission gear is configured to, when the output gear rotates in a state where the transmission gear is located at the first position, not be coupled to the output gear.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority from Japanese Patent Application No. 2021-192563 filed on Nov. 26, 2021. The entire content of the priority application is incorporated herein by reference.

BACKGROUND ART

(2) A one-way clutch that transmits rotational force in only one direction is conventionally known. The one-way clutch is used, for example, in a sheet conveyance device provided in an image forming apparatus.

DESCRIPTION

(3) In a sheet conveyance device, a one-way clutch is incorporated in a drive train that transmits drive power from a motor to a fuser (fixing device) that conveys a sheet and fixes an image on the sheet.

(4) In such a fuser, a sheet jam may occur between fixing rollers. When a jam occurs, it is

necessary to perform a jam removal process of pulling out the jammed sheet by hand. At this time, if the fixing roller and the motor are connected to each other during the jam removal process, the load of the motor and the drive train is applied to rotation of the fixing roller, which may hinder a smooth jam removal process.

(5) In order to deal with the above problem, it is conceivable to provide a driving force transmission release mechanism in addition to a one-way clutch, in driving force transmission means for transmitting driving force from a drive source to the fixing roller. The driving force transmission release mechanism performs transmission and release of driving force between the fixing roller and the drive source, thereby releasing coupling between the fixing roller and the drive source during a jam removal process. However, if such a driving force transmission release means is provided, the configuration of the driving force transmission means may become complicated.

(6) In view of the foregoing, an example of an object of this disclosure is to provide a sheet conveyance device and an image forming apparatus configured to enable a smooth jam removal process without providing a complicated mechanism in a driving force transmission means.

(7) According to one aspect, this specification discloses a sheet conveyance device. The sheet conveyance device includes a drive source, a roller configured to convey a sheet, and a one-way clutch configured to transmit a driving force from the drive source to the roller. The one-way clutch includes a fixed shaft, an input gear, an output gear, and a transmission gear. The fixed shaft extends in an axial direction. The input gear is rotatably supported by the fixed shaft. The input gear is rotatable by receiving the driving force from the drive source. The output gear is rotatably supported by the fixed shaft. The output gear is coupled to the roller and configured to transmit the driving force to the roller. The transmission gear is supported by the fixed shaft. The transmission gear is movable in the axial direction between a first position and a second position in a state where the transmission gear engages with the input gear. The first position is a position at which the transmission gear is separated from the output gear. The second position is a position at which the transmission gear is coupled to the output gear. The input gear is configured to: by rotating in a first direction by receiving the driving force from the drive source, move the transmission gear from the first position to the second position and transmit the driving force in the first direction to the output gear via the transmission gear; and by rotating in a second direction opposite the first direction by receiving the driving force from the drive source, move the transmission gear from the second position to the first position and cancel transmission of the driving force to the output gear. The transmission gear is configured to, when the output gear rotates in a state where the transmission gear is located at the first position, not be coupled to the output gear. According to another aspect, this specification also discloses an image forming apparatus including a print engine and the above-described sheet conveyance device.

(8) According to the above-described configuration, when the output gear rotates in a state where the transmission gear is located at the first position, the transmission gear is not coupled to the output gear. Because the transmission gear and the output gear are not coupled, the roller is easily rotated when performing a jam removal process of removing a jammed sheet without providing a complicated mechanism in driving force transmission means for transmitting the driving force from the drive source to the roller. Thus, the jam removal process is performed smoothly.

Description

- (1) FIG. 1 is a central cross-sectional view showing an image forming apparatus.
- (2) FIG. 2 is a side cross-sectional view showing a fuser.
- (3) FIG. 3 is a perspective view showing a heating unit and a pressure roller of the fuser;
- (4) FIG. 4 is a side view showing a driving force transmission train.
- (5) FIG. 5 is an exploded perspective view showing a one-way clutch.

- (6) FIG. 6A is a side cross-sectional view showing the one-way clutch in a state where a transmission gear is located at a first position.
- (7) FIG. 6B is a side cross-sectional view showing the one-way clutch in a state where the transmission gear is located at a second position.
- (8) FIG. 7 is a rear view showing a fixed shaft of the one-way clutch.
- (9) FIGS. 8A and 8B are perspective views showing an input gear of the one-way clutch.
- (10) FIG. 9 is an exploded perspective view of the one-way clutch in which an inclined engagement surface of the input gear is visibly shown.
- (11) FIGS. 10A and 10B are perspective views showing a transmission gear of the one-way clutch.
- (12) FIGS. 11A and 11B are perspective views showing an output gear of the one-way clutch.
- (13) FIG. 12 is a block diagram showing a controller to which a motor and a sheet discharge sensor are connected.
- (14) Next, a mode for carrying out the present disclosure will be described with reference to the accompanying drawings.
- (15) [Image Forming Apparatus]
- (16) An image forming apparatus 1 according to an embodiment shown in FIG. 1 is provided with a sheet conveyance device. For example, the image forming apparatus 1 is a laser printer that forms an image on a sheet S by an electrophotographic method.
- (17) In the following description, the right side in FIG. 1 is defined as the front side of the image forming apparatus 1, the left side in FIG. 1 is defined as the rear side of the image forming apparatus 1, the near side in the direction perpendicular to the drawing sheet of FIG. 1 is defined as the left side of the image forming apparatus 1, and the far side in the direction perpendicular to the drawing sheet of FIG. 1 is defined as the right side of the image forming apparatus 1. Further, the upper side and the lower side in FIG. 1 are defined as the upper side and the lower side of the image forming apparatus 1, respectively.
- (18) The image forming apparatus 1 includes an apparatus main body 2, a sheet feeder 3, a print engine (image forming section) 5, a fuser (fixing device) 6, a sheet discharge section 7, and a sheet conveyance device T.
- (19) The apparatus main body 2 accommodates the sheet feeder 3, the print engine 5, the fuser 6, the sheet discharge section 7, and the sheet conveyance device T.
- (20) The sheet feeder 3 includes a sheet feed tray 10 that supports the sheet S, a sheet conveyor 30, a conveyance roller pair 34, and a registration roller pair 35. The sheet feeder 3 is arranged in the lower part of the apparatus main body 2 and conveys the sheet S supported by the sheet feed tray 10 to the print engine 5. The image forming apparatus 1 has a conveyance path P for the sheet S from the sheet feeder 3 to the sheet discharge section 7 via the print engine 5.
- (21) The sheet feed tray 10 has a pressure plate 12 and a pressing plate 13. The pressure plate 12 is a plate member that supports the sheet S from below. The pressure plate 12 is rotatable about a pivot point 12a at the rear end thereof, and is movable up and down between a lowered position and a raised position by pivoting about the pivot point 12a. The pressing plate 13 is located below the pressure plate 12, and causes the pressure plate 12 to move up and down between the lowered position and the raised position.
- (22) The sheet conveyor 30 is a conveyance mechanism that separates and picks up the sheets S supported by the sheet feed tray 10 one sheet at a time and conveys the sheet S toward the print engine 5. The sheet conveyor 30 includes a sheet feed roller 31, a separation roller 32, and a separation pad 33.
- (23) The sheet feed roller 31 is a roller for sending the sheet S supported by the sheet feed tray 10 toward the separation roller 32. The separation roller 32 is arranged downstream of the sheet feed roller 31 in a sheet conveyance direction. The separation pad 33 is arranged to face the separation roller 32 and is urged toward the separation roller 32.
- (24) The sheets S sent out toward the separation roller 32 by the sheet feed roller 31 are separated

to one sheet between the separation roller **32** and the separation pad **33**. The sheet **S** separated to one sheet is sent to a conveyance path **P**.

(25) The sheet **S** sent to the conveyance path **P** is conveyed toward the print engine **5** by the conveyance roller pair **34** and the registration roller pair **35**. The registration roller pair **35** regulates the movement of the leading edge of the conveyed sheet **S** to temporarily stop the sheet **S**, and then conveys the sheet **S** toward the print engine **5** at a particular timing.

(26) The print engine **5** is arranged downstream of the sheet feeder **3** in the sheet conveyance direction, and forms an image on the sheet **S** conveyed from the sheet feeder **3**. The print engine **5** includes a process cartridge **50** that transfers an image onto the surface of the sheet **S** conveyed from the sheet feeder **3** and an exposure unit **56** that exposes the surface of a photosensitive drum **54** in the process cartridge **50**.

(27) The process cartridge **50** is arranged above the sheet feeder **3** in the apparatus main body **2**, and includes a developer storage chamber **51**, a supply roller **52**, a development roller **53**, the photosensitive drum **54**, a transfer roller **55**, and so on.

(28) The exposure unit **56** includes a laser diode, a polygon mirror, a lens, a reflecting mirror, and so on. The exposure unit **56** irradiates the photosensitive drum **54** with laser light based on image data inputted to the image forming apparatus **1**, thereby exposing the surface of the photosensitive drum **54**.

(29) The developer storage chamber **51** stores toner serving as a developer. The toner stored in the developer storage chamber **51** is sent to the supply roller **52** while being stirred by an agitator (not shown). The supply roller **52** further supplies the toner sent from the developer storage chamber **51** to the development roller **53**.

(30) The development roller **53** is arranged in close contact with the supply roller **52**, and bears toner supplied from the supply roller **52** and positively charged by a sliding contact member (not shown). A developing bias is applied to the development roller **53** by bias applying means (not shown).

(31) The photosensitive drum **54** is arranged adjacent to the development roller **53**. The surface of the photosensitive drum **54** is uniformly charged by a charger (not shown) and then exposed by the exposure unit **56**. The exposed portion of the photosensitive drum **54** has a lower potential than the other portions, and an electrostatic latent image is formed on the photosensitive drum **54** based on the image data. Then, positively charged toner is supplied from the development roller **53** to the surface of the photosensitive drum **54** on which the electrostatic latent image is formed, whereby the electrostatic latent image is visualized to form a toner image.

(32) The transfer roller **55** is arranged to face the photosensitive drum **54**. A transfer bias is applied to the transfer roller **55** by bias application means (not shown). When the sheet **S** is nipped and conveyed between the transfer roller **55** and the photosensitive drum **54** on which the toner image is formed in a state where the transfer bias is applied to the surface of the transfer roller **55**, the toner image formed on the surface of the photosensitive drum **54** is transferred to the surface of the sheet **S**. The sheet **S** on which the toner image has been transferred is conveyed to the fuser **6**.

(33) The fuser **6** has a heating unit **61** and a pressure roller **62**, and fixes the image transferred to the sheet **S** by the process cartridge **50**. The heating unit **61** is heated by being supplied with power from a power source (not shown). The pressure roller **62** is arranged to face the heating unit **61**. One of the heating unit **61** and the pressure roller **62** is urged against the other by an urging mechanism (not shown), and the heating unit **61** and the pressure roller **62** are in close contact with each other.

(34) When the sheet **S** on which the toner image has been transferred is conveyed to the fuser **6**, the fuser **6** conveys and heats the sheet **S** while nipping the sheet **S** between the heating unit **61** and the pressure roller **62**, thereby fixing the toner image on the sheet **S**. In this way, the fuser **6** conveys and heats the sheet **S** sent from the print engine **5**.

(35) The sheet discharging section **7** is located downstream of the print engine **5** in the sheet

conveyance direction, and discharges the sheet S on which an image has been formed by the print engine 5 to the outside of the image forming apparatus 1. The sheet discharge section 7 includes a sheet discharge roller pair 71 and a sheet discharge tray 72. The sheet discharge roller pair 71 is configured to discharge the sheet S conveyed along the conveyance path P from the fuser 6 toward the outside of the apparatus main body 2. The sheet discharge tray 72 is formed on the upper surface of the apparatus main body 2, and supports the sheet S discharged to the outside of the apparatus main body 2 by the sheet discharge roller pair 71.

(36) [Fuser]

(37) As shown in FIGS. 2 and 3, the heating unit 61 of the fuser 6 includes a heater 611, a holder 612, a stay 613, and a belt 614. The heater 611 is a flat plate-shaped heater extending in the left-right direction. The heater 611 has a first surface 611A and a second surface 611B opposite to the first surface 611A. The first surface 611A is supported by the holder 612.

(38) The holder 612 is made of, for example, a resin member, and has a guide surface 612a and a support wall 612b. The guide surface 612 is in contact with an inner peripheral surface 614a of the belt 614 and guides the belt 614. The support wall 612b has a support surface 612A that supports the heater 611. The support surface 612A of the support wall 612b is in contact with the first surface 611A of the heater 611. The stay 613 is a member that supports the holder 612. The stay 613 is formed by bending a plate material having greater rigidity than the holder 612, such as a steel plate, into a substantially U-shaped cross section.

(39) The belt 614 is an endless belt having heat resistance and flexibility. The belt 614 has a metal tube made of metal such as stainless steel, and a fluoro-resin layer covering the metal tube. The heater 611, the holder 612, and the stay 613 are arranged inside the belt 614. The belt 614 is configured to rotate around the heater 611, the holder 612, and the stay 613. An inner peripheral surface 614a of the belt 614 is in contact with the heater 611.

(40) The pressure roller 62 has a metal shaft 62A and an elastic layer 62B covering the shaft 62A. The pressure roller 62 is pressed against the heater 611 via the belt 614. The pressure roller 62 forms a nip portion NP for heating and pressing the sheet S by sandwiching the belt 614 with the heater 611. That is, the pressure roller 62 heats and presses the sheet S together with the heater 611 at the nip portion NP.

(41) The pressure roller 62 is configured to be rotationally driven by transmission of driving force from a motor 4 provided in the image forming apparatus 1. Rotationally driving the pressure roller 62 causes the belt 614 to rotate followingly due to the frictional force between the pressure roller 62 and the belt 614 or the sheet S sandwiched at the nip portion NP. In this way, the sheet S on which a toner image has been transferred is conveyed between the pressure roller 62 and the heated belt 614, and the toner image is thermally fixed. The pressure roller 62 is an example of a roller that conveys a sheet.

(42) [Sheet Conveyance Device]

(43) The sheet conveyance device T includes the motor 4, a driving force transmission train 8, the pressure roller 62, and so on, and conveys the sheet S. The motor 4 is an example of a drive source.

(44) [Driving Force Transmission Train]

(45) As shown in FIG. 4, the driving force transmission train 8 is a driving force transmission train for transmitting the driving force from the motor 4 to the pressure roller 62 of the fuser 6. The driving force transmission train 8 is arranged at the left end of the apparatus main body 2. The driving force transmission train 8 includes a motor gear 81, a first idle gear 82, a second idle gear 83, a third idle gear 84, a one-way clutch 90, a fourth idle gear 85, and a pressure roller gear 86.

(46) The motor gear 81 is fixed to an output shaft 4a of the motor 4, and the first idle gear 82 engages with the motor gear 81. The second idle gear 83 engages with the first idle gear 82, and the third idle gear 84 engages with the second idle gear 83. The input side of the one-way clutch 90 is connected to the third idle gear 84. The fourth idle gear 85 is connected to the output side of the one-way clutch 90. The pressure roller gear 86 engages with the fourth idle gear 85. The pressure

roller gear **86** is fixed to the shaft **62A** of the pressure roller **62** so as to be integrally rotatable.

(47) [One-Way Clutch]

(48) As shown in FIGS. 5, 6A and 6B, the one-way clutch **90** is configured to transmit the driving force from the motor **4** to the pressure roller **62**. The one-way clutch **90** includes a fixed shaft **91**, an input gear **92**, a transmission gear **93**, and an output gear **94**. The input gear **92** is the input-side gear of the one-way clutch **90**, and the output gear **94** is the output-side gear of the one-way clutch **90**.

(49) In the one-way clutch **90**, the input gear **92**, the transmission gear **93**, and the output gear **94** are arranged in this order from left to right. That is, the transmission gear **93** is located between the input gear **92** and the output gear **94** in an axial direction X.

(50) As shown in FIGS. 5 to 7, the fixed shaft **91** is a long shaft member extending in the left-right direction, and has a first shaft portion **911** and a second shaft portion **912**. The direction along the left-right direction in which the fixed shaft **91** extends is the axial direction X of the fixed shaft **91**. The first shaft portion **911** has a first outer diameter D1, and the second shaft portion **912** has a second outer diameter D2 smaller than the first outer diameter D1 ($D1 > D2$). The second shaft portion **912** is formed continuously to the right of the first shaft portion **911**.

(51) As shown in FIGS. 5, 6A, 6B, and 8, the input gear **92** is rotatably supported by the first shaft portion **911** of the fixed shaft **91** and is rotatable by receiving the driving force from the motor **4**. The input gear **92** is rotatable in a first direction W1 and a second direction W2 opposite to the first direction W1. The input gear **92** has a body **920**, an input tooth portion **921**, a transmission portion **922** (transmission protrusion), an inclined cam surface **923**, and an inclined engagement surface **924**.

(52) The body **920** is formed of a cylindrical member having a bottom portion **920a**, and the bottom portion **920a** is located at the left end of the body **920**. The input tooth portion **921** is formed in a cylindrical shape protruding leftward from the bottom portion **920a** of the body **920**, and input teeth **921a** are formed on the outer peripheral surface of the input tooth portion **921**. The input teeth **921a** engage with the third idle gear **84**. A driving force from the motor **4** is inputted to the input teeth **921a** via the motor gear **81**, the first idle gear **82**, the second idle gear **83**, and the third idle gear **84**.

(53) The transmission portion **922** is a protrusion (protruding piece) that protrudes radially inward from the inner peripheral surface of the body **920**, and is configured to engage with the transmission gear **93**. In this embodiment, the transmission portions **922** are formed at two positions. The two transmission portions **922** are located at such positions that the distance from one transmission portion **922** to the other transmission portion **922** in a circumferential direction is equal to the distance from the other transmission portion **922** to the one transmission portion **922** in the circumferential direction.

(54) The inclined cam surface **923** is an inclined surface formed on the bottom portion **920a** of the body **920** and facing the output gear **94** side in the axial direction X. Here, a phrase “the inclined cam surface **923** is an inclined surface facing the output gear **94** side” means that the inclined cam surface **923** is an inclined surface that is visible when viewed from the output gear **94** side in the axial direction X, that is, from the right side.

(55) The inclined cam surface **923** extends along the circumferential direction. The inclined cam surface **923** is inclined relative to the direction perpendicular to the axial direction X such that the upstream side of the inclined cam surface **923** in the first direction W1 is closer to the output gear **94** in the axial direction X than the downstream side. The inclined cam surface **923** has a downstream end **923a** in the first direction W1 and an upstream end **923b** in the first direction W1. In the axial direction X, the upstream end **923b** is located closer to the output gear **94** than the downstream end **923a** is.

(56) The inclined engagement surface **924** is formed inside the body **920**, and located upstream of the transmission portion **922** in the first direction W1. The inclined engagement surface **924** is an

inclined surface facing away from the output gear **94** in the axial direction X. As shown in FIG. 9, a phrase “the inclined engagement surface **924** is an inclined surface facing away from the output gear **94**” means that the inclined engagement surface **924** is an inclined surface that is visible when viewed from the opposite side of the output gear **94** in the axial direction X, that is, from the left side (see the inclined engagement surface **924** surrounded by a dotted line in FIG. 9). Note that in FIG. 9, the bottom portion **920a** of the input gear **92** is shown to be transparent such that the inclined engagement surface **924** is visible.

(57) The inclined engagement surface **924** extends along the circumferential direction. The inclined engagement surface **924** is inclined relative to the direction perpendicular to the axial direction X such that the downstream side of the inclined engagement surface **924** in the second direction W2 is closer to the output gear **94** in the axial direction X than the upstream side. In this embodiment, the inclined engagement surfaces **924** are formed at two positions. The two inclined engagement surfaces **924** are arranged at such positions that the distance from one inclined engagement surface **924** to the other inclined engagement surface **924** in a circumferential direction is equal to the distance from the other inclined engagement surface **924** to the one inclined engagement surface **924** in the circumferential direction.

(58) As shown in FIGS. 5, 6A, 6B, 10A and 10B, the transmission gear **93** is supported by the fixed shaft **91** so as to be movable in the axial direction X. The transmission gear **93** is rotatable in the first direction W1 and the second direction W2. The transmission gear **93** has a cylindrical portion **930**, a transmitted portion **931** (transmitted protrusion), a contact surface **932**, a ratchet pawl **933**, and an engagement pawl **934**.

(59) The cylindrical portion **930** is formed in a cylindrical shape. The cylindrical portion **930** has a small diameter portion **930a** at the end on the output gear **94** side in the axial direction X, the diameter of which is smaller than that of other portions. The transmitted portion **931** is a protrusion (protruding piece) that protrudes radially outward from the outer peripheral surface of the cylindrical portion **930**.

(60) In this embodiment, the transmitted portions **931** are formed at two positions. The two transmitted portions **931** are arranged at such positions that the distance from one transmitted portion **931** to the other transmitted portion **931** in a circumferential direction is equal to the distance from the other transmitted portion **931** to the one transmitted portion **931** in the circumferential direction. The transmitted portion **931** is configured to engage with the transmission portion **922** of the input gear **92** when the input gear **92** rotates in the first direction W1.

(61) The contact surface **932** is located at an end of the transmitted portion **931** facing the input gear **92** side in the axial direction X. The contact surface **932** is configured to contact the inclined cam surface **923** of the input gear **92** when the input gear **92** rotates in the first direction W1.

(62) The ratchet pawl **933** protrudes toward the output gear **94** side from an end of the cylindrical portion **930** at the output gear **94** side in the axial direction X. The ratchet pawl **933** has an engagement surface **933a** and a sliding surface **933b** formed continuously with the engagement surface **933a** in the circumferential direction. A plurality of ratchet pawls **933** are formed along the circumferential direction, and the engagement surfaces **933a** and the sliding surfaces **933b** are alternately arranged in the circumferential direction.

(63) The engagement pawl **934** protrudes from the transmitted portion **931** toward the downstream side in the first direction W1 in the circumferential direction. In this embodiment, the engagement pawls **934** are formed at two positions. The two engagement pawls **934** are arranged at such positions that the distance from one engagement pawl **934** to the other engagement pawl **934** in a circumferential direction is equal to the distance from the other engagement pawl **934** to the one engagement pawl **934** in the circumferential direction. The engagement pawl **934** is configured to engage with the inclined engagement surface **924** of the input gear **92** when the input gear **92** rotates in the second direction W2.

(64) In a state where the transmission gear **93** is engaged with the input gear **92**, the transmission gear **93** is movable in the axial direction X between a first position where the small diameter portion **930a** of the cylindrical portion **930** is supported by the first shaft portion **911** of the fixed shaft **91** (the position shown in FIG. 6A) and a second position where the small diameter portion **930a** of the cylindrical portion **930** is supported by the second shaft portion **912** of the fixed shaft **91** (the position shown in FIG. 6B).

(65) The small diameter portion **930a** of the transmission gear **93** located at the first position is supported by the first shaft portion **911** with a first sliding resistance between the small diameter portion **930a** and the first shaft portion **911**. The small diameter portion **930a** of the transmission gear **93** located at the second position is supported by the second shaft portion **912** with a second sliding resistance between the small diameter portion **930a** and the second shaft portion **912**. The second sliding resistance is smaller than the first sliding resistance.

(66) In other words, the first shaft portion **911** of the fixed shaft **91** supports the transmission gear **93** located at the first position in a state where the first sliding resistance of a particular magnitude acts between the fixed shaft **91** and the transmission gear **93**. The second shaft portion **912** of the fixed shaft **91** supports the transmission gear **93** located at the second position in a state where the second sliding resistance smaller than the first sliding resistance acts between the fixed shaft **91** and the transmission gear **93**.

(67) As shown in FIGS. 5, 6A, 6B, 11A and 11B, the output gear **94** is rotatably supported by the second shaft portion **912** of the fixed shaft **91**. The output gear **94** is connected to the pressure roller **62** of the fuser **6**, and transmits the driving force from the motor **4** to the pressure roller **62**. The output gear **94** is rotatable in the first direction W1 and the second direction W2. The output gear **94** has a body **940**, ratchet recesses **941**, output teeth **942**, and spacers **943**.

(68) The body **940** is formed in a cylindrical shape. The ratchet recess **941** is located at an end of the body **940** facing the input gear **92** in the axial direction X. The ratchet recess **941** is configured to engage with the ratchet pawl **933** of the transmission gear **93**. Specifically, the ratchet recess **941** engages with the ratchet pawl **933** when the transmission gear **93** rotates in the first direction W1 relative to the output gear **94**, and does not engage with the ratchet pawl **933** when the transmission gear **93** rotates in the second direction W2 relative to the output gear **94**.

(69) The ratchet recess **941** has an engagement surface **941a** and a sliding surface **941b** formed continuously with the engagement surface **941a** in the circumferential direction. A plurality of ratchet recesses **941** are formed along the circumferential direction, and the engagement surfaces **941a** and the sliding surfaces **941b** are alternately arranged in the circumferential direction.

(70) The ratchet recess **941** is separated from the ratchet pawl **933** of the transmission gear **93** in the axial direction X when the transmission gear **93** is at the first position. The ratchet recess **941** is coupled to the ratchet pawl **933** of the transmission gear **93** in the axial direction X when the transmission gear **93** is at the second position.

(71) When the transmission gear **93** rotates in the first direction W1 relative to the output gear **94** in a state where the transmission gear **93** is at the second position, the engagement surface **933a** of the ratchet pawl **933** and the engagement surface **941a** of the ratchet recess **941** are engaged and the transmission gear **93** and the output gear **94** rotate together.

(72) When the transmission gear **93** rotates in the second direction W2 relative to the output gear **94** in a state where the transmission gear **93** is at the second position, the sliding surface **933b** of the ratchet pawl **933** slides relative to the sliding surface **941b** of the ratchet recess **941** in the circumferential direction, and the engagement surface **933a** and the engagement surface **941a** are disengaged. In this case, even if the transmission gear **93** rotates in the second direction W2, the output gear **94** does not rotate.

(73) When the transmission gear **93** is at the first position, the ratchet recess **941** and the ratchet pawl **933** are separated from each other. Thus, the output gear **94** neither rotates when the transmission gear **93** rotates in the first direction W1 nor when the transmission gear **93** rotates in

the second direction W2. When the output gear **94** rotates in a state where the transmission gear **93** is at the first position, the transmission gear **93** is not coupled to the output gear **94**.

(74) The output teeth **942** are formed on the outer peripheral surface of the body **940** and engages with the fourth idle gear **85**. The output teeth **942** are coupled to the pressure roller **62** of the fuser **6** via the fourth idle gear **85** and pressure roller gear **86**. The output teeth **942** transmit the driving force from the motor **4** to the pressure roller **62** via the fourth idle gear **85** and pressure roller gear **86**.

(75) The spacer **943** protrudes toward the input gear **92** from an end of the body **940** facing the input gear **92** side in the axial direction X, and is in contact with the input gear **92**. A plurality of spacers **943** are formed at intervals along the circumferential direction at the outer peripheral edge of the body **940**.

(76) The spacer **943** regulates the position of the output gear **94** relative to the input gear **92** in the axial direction X by contacting the input gear **92**. By regulating the position of the output gear **94** relative to the input gear **92** with the spacer **943**, the transmission gear **93** is movable in the axial direction X between the input gear **92** and the output gear **94**.

(77) [Operation of One-Way Clutch]

(78) Next, the operation of the one-way clutch **90** will be described.

(79) In the one-way clutch **90** shown in FIG. 6A, the transmission gear **93** is located at the first position, and the contact surface **932** of the transmission gear **93** is in contact with the downstream end **923a** of the inclined cam surface **923** of the input gear **92**. The transmission gear **93** located at the first position is separated from the output gear **94** in the axial direction X.

(80) In a state where the transmission gear **93** is located at the first position in this way, even if the output gear **94** rotates in the first direction W1 or the second direction W2, the output gear **94** and the transmission gear **93** are not coupled to each other. Thus, in a state where the transmission gear **93** is located at the first position, even if the pressure roller **62** coupled to the output gear **94** rotates, the pressure roller **62** is not coupled to the motor **4** which is coupled to the input gear **92**.

(81) When the transmission gear **93** is located at the first position, the small diameter portion **930a** of the cylindrical portion **930** of the transmission gear **93** is supported by the first shaft portion **911** of the fixed shaft **91** in a state where the first sliding resistance acts therebetween.

(82) When the input gear **92** receives the driving force from the motor **4** and rotates in the first direction W1 in the state shown in FIG. 6A, the inclined cam surface **923** in contact with the contact surface **932** of the transmission gear **93** moves in the circumferential direction, and the transmission gear **93** receives force in the circumferential direction from the inclined cam surface **923** due to a sliding resistance between the transmission gear **93** and the inclined cam surface **923**.

(83) The transmission gear **93** is supported by the first shaft portion **911** in a state where the first sliding resistance acts between the small diameter portion **930a** and the first shaft portion **911**. The first sliding resistance is set to be greater than the sliding resistance in the circumferential direction generated between the transmission gear **93** and the inclined cam surface **923**. Thus, when the input gear **92** rotates in the first direction W1 in a state where the contact surface **932** of the transmission gear **93** and the inclined cam surface **923** are in contact with each other, the transmission gear **93** keeps a state where rotation relative to the fixed shaft **91** is suppressed, and thus does not rotate at the same speed together with the input gear **92**.

(84) As a result, the phase of the transmission gear **93** relative to the inclined cam surface **923** in the first direction W1 is shifted, and the position of the inclined cam surface **923** contacting the contact surface **932** of the transmission gear **93** is shifted from the downstream end **923a** to the upstream side in the first direction W1. The upstream side of the inclined cam surface **923** in the first direction W1 is located closer to the output gear **94** than the downstream side of the inclined cam surface **923** is. Thus, when the position of the inclined cam surface **923** contacting the contact surface **932** of the transmission gear **93** moves to the upstream side in the first direction W1, the transmission gear **93** is pressed by the inclined cam surface **923** toward the output gear **94** in the

axial direction X and moves.

(85) As shown in FIG. 6B, when the position of the inclined cam surface **923** contacting the contact surface **932** of the transmission gear **93** reaches the upstream end **923b**, the transmission gear **93** moves to the second position.

(86) In this way, when the input gear **92** rotates in the first direction **W1**, the first sliding resistance between the small diameter portion **930a** and the first shaft portion **911** prevents the transmission gear **93** located at the first position from rotating at the same speed together with the input gear **92**, and thus enables the transmission gear **93** to move from the first position to the second position.

(87) After the transmission gear **93** has moved to the second position, the transmission portion **922** of the input gear **92** rotating in the first direction **W1** contacts the transmitted portion **931** of the transmission gear **93**. When the transmission portion **922** contacts the transmitted portion **931**, a force in the first direction **W1** acts on the transmission gear **93** from the input gear **92**, and the input gear **92** and the transmission gear **93** rotate together in the first direction **W1**.

(88) The second sliding resistance between the second shaft portion **912** and the small diameter portion **930a** of the transmission gear **93** located at the second position is smaller than the first sliding resistance. Thus, the transmission gear **93** rotates smoothly against the second sliding resistance.

(89) In the one-way clutch **90**, the first outer diameter **D1** of the first shaft portion **911** for supporting the transmission gear **93** located at the first position is different from the second outer diameter **D2** of the second shaft portion **912** for supporting the transmission gear **93** located at the second position. This causes the magnitude of the first sliding resistance and the magnitude of the second sliding resistance be different from each other, thereby simplifying the configuration of the one-way clutch **90**.

(90) When the transmission gear **93** is located at the second position and rotates in the first direction **W1**, the engagement surface **933a** of the ratchet pawl **933** of the transmission gear **93** engages with the engagement surface **941a** of the ratchet recess **941** of the output gear **94**, and the transmission gear **93** and the output gear **94** rotate integrally in the first direction **W1**.

(91) In this way, the input gear **92** receives the driving force from the motor **4** and rotates in the first direction **W1**, thereby moving the transmission gear **93** from the first position to the second position, so that the transmission gear **93** is coupled to the output gear **94** and the driving force in the first direction **W1** from the motor **4** is transmitted to the output gear **94** via the transmission gear **93**.

(92) In other words, when the input gear **92** rotates in the first direction **W1**, the transmission gear **93** located at the second position and coupled to the output gear **94** rotates relative to the fixed shaft **91** together with the input gear **92** against the second sliding resistance. This enables the driving force to be transmitted from the transmission gear **93** to the output gear **94**.

(93) In the one-way clutch **90**, when the input gear **92** rotates in the first direction **W1**, the transmission gear **93** is pressed toward the output gear **94** by the inclined cam surface **923** and moves from the first position to the second position. This enables switching from a state where the transmission of the driving force to the output gear **94** is canceled to a state where the driving force is transmitted to the output gear **94** via the transmission gear **93** with a simple configuration.

(94) Conversely, in the one-way clutch **90**, when the input gear **92** receives the driving force from the motor **4** and rotates in the second direction **W2** from the state where the transmission gear **93** is located at the second position shown in FIG. 6B, the transmission gear **93** is moved from the second position to the first position so as to cancel transmission of the driving force to the output gear **94**.

(95) Specifically, when the input gear **92** receives the driving force from the motor **4** and rotates in the second direction **W2** in a state where the transmission gear **93** is at the second position, the inclined engagement surface **924** of the input gear **92** engages with the engagement pawl **934** of the transmission gear **93**. The downstream side of the inclined engagement surface **924** in the second

direction W2 is located closer to the output gear **94** than the upstream side of the inclined engagement surface **924** in the second direction W2. Thus, when the engagement pawl **934** engages with the inclined engagement surface **924** and is pressed in the second direction W2, the position of the inclined engagement surface **924** contacting the engagement pawl **934** moves upstream in the second direction W2, and the transmission gear **93** is pulled toward the input gear **92** in the axial direction X by the inclined engagement surface **924**.

(96) In other words, when the transmission gear **93** is located at the second position and the input gear **92** rotates in the second direction W2, the transmission gear **93** is pulled toward the input gear **92** in the axial direction X by the inclined engagement surface **924** and moves to the first position. As a result, the transmission gear **93** and the output gear **94** are separated from each other, and transmission of the driving force from the input gear **92** to the output gear **94** is cancelled.

(97) As described above, in the one-way clutch **90**, by rotating the input gear **92** in the second direction, the inclined engagement surface **924** and the engagement pawl **934** are engaged, and the transmission gear **93** is moved from the second position to the first position.

(98) In the one-way clutch **90**, when the transmission gear **93** is at the first position, the transmission gear **93** and the output gear **94** are separated without being coupled. Thus, even when the output gear **94** is rotated, the transmission gear **93** does not rotate together with the output gear **94** and the output gear **94** idles.

(99) Thus, the load of the motor **4** and the gear train between the motor **4** and the input gear **92** is not applied to the output gear **94** coupled to the pressure roller **62**. This enables the pressure roller **62** to be easily rotated when removing a jammed sheet S, without providing a complicated mechanism such as a driving force transmission canceling mechanism in addition to the one-way clutch **90** in the driving force transmission train **8**. Thus, removing the jammed sheet S is performed smoothly.

(100) When the transmission gear **93** is at the first position, the one-way clutch **90** cuts off the transmission of the driving force from the motor **4** to the pressure roller **62**. Thus, even if the pressure roller **62** is rotated when removing a jammed sheet, the load of the motor **4** is not applied to the pressure roller **62**, and a process of removing a jammed sheet is performed smoothly.

(101) Since the one-way clutch **90** is coupled to the pressure roller **62** of the fuser **6**, the sheet S is smoothly removed from the fuser **6** when a jam of the sheet S occurs in the fuser **6**.

(102) The heating unit **61** of the fuser **6** has the heater **611** of a flat plate shape, and the pressure roller **62** is pressed against the heater **611**. Thus, the rotational torque of the pressure roller **62** tends to be larger than a case where the heating unit **61** is formed by a roller member. However, in the one-way clutch **90**, by moving the transmission gear **93** to the first position where the transmission gear **93** and the output gear **94** are separated from each other, the load applied to the pressure roller **62** does not increase and a jammed sheet is removed smoothly.

(103) [Motor Drive Control When a Jam Occurs]

(104) As shown in FIG. **12**, the image forming apparatus **1** includes a controller C and the sheet discharge sensor **98**. The controller C is connected to the motor **4** and configured to control the operation of the motor **4**. As shown in FIG. **1**, the sheet discharge sensor **98** is arranged between the fuser **6** and the sheet discharge roller pair **71** in the sheet conveyance direction, and is configured to detect the sheet S that passes through the fuser **6**.

(105) The controller C is configured to, when the detection of the sheet S by the sheet discharge sensor **98** continues for a particular time or longer, for example, determine that the sheet S is jammed in the fuser **6**. In response to determining that the sheet S is jammed, the controller C controls the motor **4** to rotate in the direction in which the input gear **92** rotates in the second direction W2.

(106) When the motor **4** is driven to rotate the input gear **92** in the second direction W2, the inclined engagement surface **924** engages with the engagement pawl **934**, the transmission gear **93** moves to the first position, and the transmission gear **93** separates from the output gear **94**.

(107) In this way, when the sheet S is jammed, the motor 4 is driven under the control of the controller C to rotate the input gear 92 in the second direction W2, thereby moving the transmission gear 93 from the second position to the first position so that the transmission gear 93 and the output gear 94 are separated from each other. Thus, when the pressure roller 62 is rotated for removing a jam, the load of the motor 4 is not applied to the pressure roller 62, and the jam is removed smoothly.

(108) While the invention has been described in conjunction with various example structures outlined above and illustrated in the figures, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or that may be presently unforeseen, may become apparent to those having at least ordinary skill in the art. Accordingly, the example embodiments of the disclosure, as set forth above, are intended to be illustrative of the invention, and not limiting the invention. Various changes may be made without departing from the spirit and scope of the disclosure. Thus, the disclosure is intended to embrace all known or later developed alternatives, modifications, variations, improvements, and/or substantial equivalents. Some specific examples of potential alternatives, modifications, or variations in the described invention are provided below:

(109) For example, in response to determining that a sheet S is jammed at a position other than the fuser 6, the controller C may control the motor 4 to rotate in a direction in which the input gear 92 rotates in the second direction W2.

Claims

1. A sheet conveyance device comprising: a motor; a roller configured to convey a sheet; and a one-way clutch configured to transmit a driving force from the motor to the roller, the one-way clutch including: a fixed shaft extending in an axial direction; an input gear rotatably supported by the fixed shaft, the input gear being rotatable by receiving the driving force from the motor; an output gear rotatably supported by the fixed shaft, the output gear being coupled to the roller and configured to transmit the driving force to the roller; and a transmission gear supported by the fixed shaft, the transmission gear being movable in the axial direction between a first position and a second position in a state where the transmission gear engages with the input gear, the first position being a position at which the transmission gear is separated from the output gear, the second position being a position at which the transmission gear is coupled to the output gear, the input gear being configured to: by rotating in a first direction by receiving the driving force from the motor, move the transmission gear from the first position to the second position and transmit the driving force in the first direction to the output gear via the transmission gear; and by rotating in a second direction opposite the first direction by receiving the driving force from the motor, move the transmission gear from the second position to the first position and cancel transmission of the driving force to the output gear, the transmission gear being configured to, when the output gear rotates in a state where the transmission gear is located at the first position, not be coupled to the output gear, wherein the fixed shaft is configured to: support the transmission gear located at the first position in a state where a first sliding resistance of a particular magnitude acts between the fixed shaft and the transmission gear; and support the transmission gear located at the second position in a state where a second sliding resistance acts between the fixed shaft and the transmission gear, the second sliding resistance being smaller than the first sliding resistance.
2. The sheet conveyance device according to claim 1, wherein, when the roller rotates in a state where the transmission gear is located at the first position, the roller is not coupled to the motor.
3. The sheet conveyance device according to claim 1, wherein the transmission gear located at the first position is configured to: when the input gear rotates in the first direction, be moved by the input gear from the first position to the second position by keeping a state where rotation relative to the fixed shaft is suppressed due to the first sliding resistance; and wherein the transmission gear

located at the second position is configured to: when the input gear rotates in the first direction, rotate relative to the fixed shaft together with the input gear against the second sliding resistance.

4. The sheet conveyance device according to claim 1, wherein the fixed shaft includes: a first shaft portion having a first outer diameter; and a second shaft portion having a second outer diameter smaller than the first outer diameter; wherein the first shaft portion supports the transmission gear located at the first position in a state where the first sliding resistance acts between the first shaft portion and the transmission gear; and wherein the second shaft portion supports the transmission gear located at the second position in a state where the second sliding resistance acts between the second shaft portion and the transmission gear.

5. The sheet conveyance device according to claim 1, wherein the transmission gear is located between the input gear and the output gear in the axial direction; wherein the input gear includes: input teeth to which the driving force is inputted; a transmission protrusion engageable with the transmission gear; and an inclined cam surface facing toward the output gear in the axial direction, the inclined cam surface extending along a circumferential direction, an upstream side of the inclined cam surface in the first direction being closer to the output gear in the axial direction than a downstream side of the inclined cam surface in the first direction is; wherein the transmission gear includes: a transmitted protrusion engageable with the transmission protrusion when the input gear rotates in the first direction; a contact surface configured to contact the inclined cam surface when the input gear rotates in the first direction; and a ratchet pawl protruding toward the output gear in the axial direction; and wherein the output gear includes: a ratchet recess configured to: engage with the ratchet pawl when the transmission gear rotates in the first direction; and not engage with the ratchet pawl when the transmission gear rotates in the second direction; and output teeth coupled to the roller.

6. The sheet conveyance device according to claim 5, wherein the input gear includes: an inclined engagement surface facing away from the output gear in the axial direction, the inclined engagement surface extending in a circumferential direction, a downstream side of the inclined engagement surface in the second direction being closer to the output gear in the axial direction than an upstream side of the inclined engagement surface in the second direction is; and wherein the transmission gear includes: an engagement pawl configured to engage with the inclined engagement surface when the input gear rotates in the second direction.

7. The sheet conveyance device according to claim 1, further comprising a controller configured to, in response to determining that a jam of a sheet occurs, control the motor to drive in a direction in which the input gear rotates in the second direction.

8. The sheet conveyance device according to claim 1, wherein the output gear includes a body and a spacer, the spacer protruding from the body toward the input gear in the axial direction, the spacer being in contact with the input gear to regulate a position of the output gear relative to the input gear in the axial direction; and wherein the transmission gear is located between the input gear and the output gear in the axial direction, the transmission gear being movable in the axial direction between the input gear and the output gear.

9. An image forming apparatus comprising: a print engine configured to form an image on a sheet; a fuser configured to convey and heat the sheet conveyed from the print engine, the fuser including a roller; a motor; and a one-way clutch configured to transmit a driving force from the motor to the roller, the one-way clutch including: a fixed shaft extending in an axial direction; an input gear rotatably supported by the fixed shaft, the input gear being rotatable by receiving the driving force from the motor; an output gear rotatably supported by the fixed shaft, the output gear being coupled to the roller and configured to transmit the driving force to the roller; and a transmission gear supported by the fixed shaft, the transmission gear being movable in the axial direction between a first position and a second position in a state where the transmission gear engages with the input gear, the first position being a position at which the transmission gear is separated from the output gear, the second position being a position at which the transmission gear is coupled to the output

gear, the input gear being configured to: by rotating in a first direction by receiving the driving force from the motor, move the transmission gear from the first position to the second position and transmit the driving force in the first direction to the output gear via the transmission gear; and by rotating in a second direction opposite the first direction by receiving the driving force from the motor, move the transmission gear from the second position to the first position and cancel transmission of the driving force to the output gear, the transmission gear being configured to, when the output gear rotates in a state where the transmission gear is located at the first position, not be coupled to the output gear, wherein the fixed shaft is configured to: support the transmission gear located at the first position in a state where a first sliding resistance of a particular magnitude acts between the fixed shaft and the transmission gear; and support the transmission gear located at the second position in a state where a second sliding resistance acts between the fixed shaft and the transmission gear, the second sliding resistance being smaller than the first sliding resistance.

10. The image forming apparatus according to claim 9, wherein the fuser includes a heater having a flat plate shape; and wherein the roller is a pressure roller configured to heat and press the sheet in cooperation with the heater.

11. The image forming apparatus according to claim 9, wherein, when the roller rotates in a state where the transmission gear is located at the first position, the roller is not coupled to the motor.

12. The image forming apparatus according to claim 9, wherein the transmission gear located at the first position is configured to: when the input gear rotates in the first direction, be moved by the input gear from the first position to the second position by keeping a state where rotation relative to the fixed shaft is suppressed due to the first sliding resistance; and wherein the transmission gear located at the second position is configured to: when the input gear rotates in the first direction, rotate relative to the fixed shaft together with the input gear against the second sliding resistance.

13. The image forming apparatus according to claim 9, wherein the fixed shaft includes: a first shaft portion having a first outer diameter; and a second shaft portion having a second outer diameter smaller than the first outer diameter; wherein the first shaft portion supports the transmission gear located at the first position in a state where the first sliding resistance acts between the first shaft portion and the transmission gear; and wherein the second shaft portion supports the transmission gear located at the second position in a state where the second sliding resistance acts between the second shaft portion and the transmission gear.

14. The image forming apparatus according to claim 9, wherein the transmission gear is located between the input gear and the output gear in the axial direction; wherein the input gear includes: input teeth to which the driving force is inputted; a transmission protrusion engageable with the transmission gear; and an inclined cam surface facing toward the output gear in the axial direction, the inclined cam surface extending along a circumferential direction, an upstream side of the inclined cam surface in the first direction being closer to the output gear in the axial direction than a downstream side of the inclined cam surface in the first direction is; wherein the transmission gear includes: a transmitted protrusion engageable with the transmission protrusion when the input gear rotates in the first direction; a contact surface configured to contact the inclined cam surface when the input gear rotates in the first direction; and a ratchet pawl protruding toward the output gear in the axial direction; and wherein the output gear includes: a ratchet recess configured to: engage with the ratchet pawl when the transmission gear rotates in the first direction; and not engage with the ratchet pawl when the transmission gear rotates in the second direction; and output teeth coupled to the roller.

15. The image forming apparatus according to claim 14, wherein the input gear includes: an inclined engagement surface facing away from the output gear in the axial direction, the inclined engagement surface extending in a circumferential direction, a downstream side of the inclined engagement surface in the second direction being closer to the output gear in the axial direction than an upstream side of the inclined engagement surface in the second direction is; and wherein the transmission gear includes: an engagement pawl configured to engage with the inclined

engagement surface when the input gear rotates in the second direction.

16. The image forming apparatus according to claim 9, further comprising a controller configured to, in response to determining that a jam of a sheet occurs, control the motor to drive in a direction in which the input gear rotates in the second direction.

17. The image forming apparatus according to claim 9, wherein the output gear includes a body and a spacer, the spacer protruding from the body toward the input gear in the axial direction, the spacer being in contact with the input gear to regulate a position of the output gear relative to the input gear in the axial direction; and wherein the transmission gear is located between the input gear and the output gear in the axial direction, the transmission gear being movable in the axial direction between the input gear and the output gear.

18. A sheet conveyance device comprising: a motor; a roller configured to convey a sheet; and a one-way clutch configured to transmit a driving force from the motor to the roller, the one-way clutch including: a fixed shaft extending in an axial direction; an input gear rotatably supported by the fixed shaft, the input gear being rotatable by receiving the driving force from the motor; an output gear rotatably supported by the fixed shaft, the output gear being coupled to the roller and configured to transmit the driving force to the roller; and a transmission gear supported by the fixed shaft, the transmission gear being movable in the axial direction between a first position and a second position in a state where the transmission gear engages with the input gear, the first position being a position at which the transmission gear is separated from the output gear, the second position being a position at which the transmission gear is coupled to the output gear, the input gear being configured to: by rotating in a first direction by receiving the driving force from the motor, move the transmission gear from the first position to the second position and transmit the driving force in the first direction to the output gear via the transmission gear; and by rotating in a second direction opposite the first direction by receiving the driving force from the motor, move the transmission gear from the second position to the first position and cancel transmission of the driving force to the output gear, the transmission gear being configured to, when the output gear rotates in a state where the transmission gear is located at the first position, not be coupled to the output gear, wherein the fixed shaft includes: a first shaft portion having a first outer diameter; and a second shaft portion having a second outer diameter smaller than the first outer diameter; wherein the first shaft portion supports the transmission gear located at the first position; and wherein the second shaft portion supports the transmission gear located at the second position.
